

SAMIK SENGUPTA

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer experienced in building production-grade batch and real-time data pipelines across the modern data stack, including Python, SQL, Apache Kafka, Apache Spark, Apache Airflow, dbt, Snowflake, and AWS S3. Designed and shipped 2 end-to-end pipeline projects implementing ETL, ELT, stream processing, data lakehouse, and data quality patterns, fully containerized with Docker. Certified in dbt Fundamentals and focused on delivering reliable, scalable, and well-tested data infrastructure.

TECHNICAL SKILLS

Programming and Querying: Python, SQL, PySpark, Pandas

Stream Processing: Apache Kafka, Spark Structured Streaming, Real-Time Data Pipelines

Batch and Orchestration: Apache Airflow, dbt (data build tool), ETL and ELT Pipelines

Data Warehousing and Storage: Snowflake, PostgreSQL, AWS S3, Data Lake, Data Warehouse

Cloud and DevOps: AWS (S3, IAM), Docker, Docker Compose, Git, GitHub

Core Concepts: Data Modeling, Data Quality Testing, Schema Design, Distributed Processing

DATA ENGINEERING PROJECTS

Real-Time Market Data Streaming Pipeline | *Python, Kafka, Spark, Snowflake, AWS S3, Docker* | [GitHub](#)

- Engineered an end-to-end real-time streaming pipeline ingesting live stock-price data for 4 stocks through the Alpha Vantage API into Apache Kafka, partitioned by symbol across 4 partitions to preserve per-stock event ordering.
- Developed a PySpark Structured Streaming consumer using a dual-sink pattern, writing each 30-second micro-batch simultaneously to 2 targets, AWS S3 (raw data lake) and Snowflake (analytics warehouse), with independent per-sink failure handling.
- Implemented exactly-once processing using Spark offset management to guarantee zero data loss and no duplicate records across pipeline restarts.
- Containerized the full stack across 5 services using a production-grade Docker Compose setup with health checks, an init container for deterministic topic creation, and dependency conditions that eliminated startup race conditions.
- Diagnosed and resolved 7 environment-level failures, including Scala and JAR version conflicts, Hadoop-AWS configuration issues, and native I/O checkpoint errors.

Batch ELT Market Data Pipeline | *Python, Airflow, dbt, PostgreSQL, Snowflake, AWS S3* | [GitHub](#)

- Built a daily-scheduled batch ELT pipeline extracting stock data through a REST API, loading raw JSON into AWS S3 and PostgreSQL, and transforming it into analytics-ready tables in Snowflake.
- Modeled transformations in dbt with layered staging and mart models, source freshness checks, and unique and not-null data quality tests, generating an automated data-lineage graph for full transparency.
- Orchestrated the workflow with Apache Airflow on a custom Docker image, using cross-task communication and operator chaining for reliable daily scheduling.
- Applied a data lakehouse architecture with a dual-write S3 raw layer enabling full reprocessing alongside a structured warehouse layer optimized for analytical queries.

EXPERIENCE

Trainee Data Engineer | Kinaxs Pvt Ltd | Remote | May 2026 - Present

- Developed 5+ Python ETL scripts and SQL queries supporting internal data extraction, transformation, and schema-design tasks across multiple datasets.
- Applied modern data engineering practices across 2 live projects, including pipeline development, data modeling, and version-controlled Git workflows, improving data reliability.

CERTIFICATIONS

dbt Fundamentals | dbt Labs | May 2026

EDUCATION

Bachelor of Computer Applications (BCA) | 2022 - 2025

Acharya Institute of Graduate Studies, Bengaluru City University | CGPA: 7.8